

## REVELSTOKE, BC, Canada

WMO: 718820

Lat: 50.9581N

Lon: 118.1765W

Elev: 445

StdP: 96.10

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -16.0      | -13.2 | -20.5                         | 0.6 | -14.1 | -17.4 | 0.9 | -12.0 | 12.8                  | -3.6 | 11.3 | -4.1 | 1.8                   | 0    | 0.561 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.2                   | 30.7            | 17.9 | 28.5 | 17.0 | 26.5 | 16.3 | 19.2                | 27.4 | 18.2 | 25.7 | 17.3 | 24.1 | 2.3                  | 320  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.6                            | 12.4 | 21.2 | 15.6 | 11.7 | 20.1 | 14.8 | 11.1 | 19.1 | 56.5          | 27.7 | 53.1 | 25.6 | 50.3 | 24.3 | 24.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |     |      | Extreme Annual Temperature |       |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |  |
|-------------------|-----|------|----------------------------|-------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|--|
| 1%                |     | 2.5% | 5%                         | Mean  |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |  |
| Min               |     | Max  |                            | Min   | Max  |                    | Min                                                | Max       | Min  | Max        | Min  | Max        | Min  | Max        |      |  |
| (a)               | (b) | (c)  |                            | (e)   | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |
| 9.7               | 7.8 | 6.3  |                            | -19.3 | 34.0 | 4.0                | 2.2                                                | -22.1     | 35.6 | -24.5      | 36.9 | -26.7      | 38.2 | -29.6      | 39.8 |  |
|                   |     |      |                            | -19.6 | 21.4 | 4.0                | 1.4                                                | -22.5     | 22.4 | -24.8      | 23.2 | -27.0      | 23.9 | -29.9      | 24.9 |  |
|                   |     |      |                            | DB    |      |                    |                                                    |           |      |            |      |            |      |            |      |  |
|                   |     |      |                            | WB    |      |                    |                                                    |           |      |            |      |            |      |            |      |  |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 7.5   | -3.1  | -2.1  | 2.0   | 7.3   | 12.8  | 16.1  | 19.1  | 18.3  | 13.4  | 6.8   | 1.5   | -2.7 | (6)  |
| (7)  |                                                                   | DBStd       | 8.72  | 4.88  | 4.13  | 3.24  | 2.87  | 3.28  | 2.92  | 3.13  | 2.89  | 2.92  | 2.82  | 3.46  | 4.43 | (7)  |
| (8)  |                                                                   | HDD10.0     | 1851  | 406   | 338   | 248   | 89    | 11    | 0     | 0     | 4     | 104   | 256   | 394   |      | (8)  |
| (9)  |                                                                   | HDD18.3     | 4051  | 664   | 571   | 506   | 329   | 172   | 79    | 28    | 37    | 149   | 356   | 506   | 653  | (9)  |
| (10) |                                                                   | CDD10.0     | 946   | 0     | 0     | 0     | 9     | 99    | 184   | 283   | 258   | 107   | 6     | 0     | 0    | (10) |
| (11) |                                                                   | CDD18.3     | 106   | 0     | 0     | 0     | 0     | 2     | 13    | 53    | 36    | 2     | 0     | 0     | 0    | (11) |
| (12) |                                                                   | CDH23.3     | 1331  | 0     | 0     | 0     | 3     | 97    | 209   | 580   | 403   | 38    | 0     | 0     | 0    | (12) |
| (13) |                                                                   | CDH26.7     | 411   | 0     | 0     | 0     | 0     | 17    | 56    | 209   | 122   | 6     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 1.7   | 1.6   | 1.6   | 1.8   | 2.0   | 2.1   | 2.0   | 1.8   | 1.6   | 1.4   | 1.4   | 1.8   | 1.9  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 1018  | 121   | 95    | 71    | 54    | 58    | 75    | 69    | 62    | 67    | 81    | 121   | 142  | (15) |
| (16) |                                                                   | PrecMax     | 1285  | 262   | 182   | 149   | 116   | 108   | 122   | 153   | 152   | 152   | 174   | 232   | 258  | (16) |
| (17) |                                                                   | PrecMin     | 801   | 36    | 23    | 20    | 8     | 21    | 35    | 14    | 6     | 10    | 17    | 24    | 61   | (17) |
| (18) |                                                                   | PrecStd     | 128   | 60    | 44    | 30    | 27    | 24    | 27    | 37    | 37    | 37    | 44    | 53    | 56   | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 4.3   | 5.9   | 13.9  | 22.6  | 28.9  | 31.3  | 34.2  | 32.4  | 27.0  | 17.3  | 9.3   | 5.0  | (19) |
| (20) |                                                                   |             | MCWB  | 2.7   | 3.1   | 7.1   | 12.0  | 15.3  | 17.5  | 18.8  | 18.5  | 16.5  | 11.1  | 7.5   | 3.6  | (20) |
| (21) |                                                                   | 2%          | DB    | 3.0   | 4.2   | 10.2  | 18.8  | 26.0  | 28.1  | 31.3  | 29.9  | 24.0  | 14.5  | 7.6   | 3.6  | (21) |
| (22) |                                                                   |             | MCWB  | 1.9   | 2.3   | 5.4   | 9.8   | 14.2  | 16.4  | 18.2  | 17.8  | 16.0  | 10.3  | 6.4   | 2.5  | (22) |
| (23) |                                                                   | 5%          | DB    | 2.2   | 3.1   | 7.9   | 16.2  | 23.6  | 25.7  | 29.0  | 27.6  | 21.5  | 12.6  | 6.6   | 2.8  | (23) |
| (24) |                                                                   |             | MCWB  | 1.3   | 1.7   | 4.3   | 8.4   | 12.8  | 15.3  | 17.4  | 17.1  | 14.7  | 9.5   | 5.6   | 1.8  | (24) |
| (25) |                                                                   | 10%         | DB    | 1.5   | 2.2   | 6.0   | 13.7  | 20.8  | 23.2  | 26.7  | 25.5  | 19.2  | 11.0  | 5.6   | 1.9  | (25) |
| (26) |                                                                   |             | MCWB  | 0.9   | 1.2   | 3.5   | 7.4   | 11.8  | 14.6  | 16.7  | 16.7  | 13.7  | 8.7   | 4.7   | 1.2  | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 3.0   | 3.8   | 7.8   | 12.4  | 16.5  | 19.3  | 20.9  | 20.4  | 18.1  | 13.1  | 8.0   | 4.2  | (27) |
| (28) |                                                                   |             | MCDWB | 4.0   | 4.9   | 12.5  | 20.8  | 26.5  | 28.1  | 29.2  | 28.5  | 24.3  | 15.2  | 8.7   | 4.6  | (28) |
| (29) |                                                                   | 2%          | WB    | 2.1   | 2.7   | 6.1   | 10.6  | 14.8  | 17.4  | 19.4  | 19.1  | 16.6  | 11.6  | 6.8   | 2.9  | (29) |
| (30) |                                                                   |             | MCDWB | 2.7   | 3.5   | 9.2   | 17.3  | 23.4  | 25.4  | 27.9  | 26.9  | 21.8  | 13.3  | 7.4   | 3.3  | (30) |
| (31) |                                                                   | 5%          | WB    | 1.5   | 2.0   | 4.9   | 9.2   | 13.7  | 16.2  | 18.4  | 18.1  | 15.5  | 10.2  | 5.9   | 2.0  | (31) |
| (32) |                                                                   |             | MCDWB | 2.0   | 2.7   | 7.1   | 14.5  | 21.3  | 23.3  | 26.3  | 25.2  | 20.0  | 11.8  | 6.5   | 2.5  | (32) |
| (33) |                                                                   | 10%         | WB    | 1.0   | 1.4   | 3.8   | 8.1   | 12.6  | 15.2  | 17.5  | 17.2  | 14.4  | 9.0   | 4.8   | 1.2  | (33) |
| (34) |                                                                   |             | MCDWB | 1.3   | 2.0   | 5.6   | 12.6  | 19.3  | 21.2  | 24.6  | 23.5  | 18.1  | 10.6  | 5.4   | 1.7  | (34) |
| (35) | Mean Daily Temperature Range                                      | 5% DB       | MDBR  | 4.0   | 5.9   | 7.5   | 11.2  | 13.1  | 11.7  | 13.2  | 12.9  | 10.6  | 6.9   | 4.0   | 3.6  | (35) |
| (36) |                                                                   |             | MCDBR | 3.4   | 5.7   | 10.8  | 16.5  | 19.2  | 17.6  | 17.8  | 17.2  | 15.0  | 9.8   | 4.8   | 3.4  | (36) |
| (37) |                                                                   |             | MCWBR | 2.7   | 4.1   | 6.5   | 8.3   | 8.6   | 7.7   | 6.9   | 7.3   | 7.8   | 5.9   | 3.4   | 2.7  | (37) |
| (38) |                                                                   | 5% WB       | MCDWR | 2.9   | 4.7   | 9.6   | 14.1  | 16.6  | 15.5  | 16.1  | 15.3  | 13.6  | 8.1   | 4.1   | 3.2  | (38) |
| (39) |                                                                   |             | MCWBR | 2.4   | 3.5   | 6.1   | 7.7   | 7.8   | 7.2   | 7.0   | 7.1   | 7.4   | 5.4   | 3.1   | 2.7  | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.275 | 0.276 | 0.303 | 0.342 | 0.378 | 0.396 | 0.370 | 0.380 | 0.351 | 0.334 | 0.289 | 0.264 | (40) |      |
| (41) |                                                                   | taud        | 2.248 | 2.286 | 2.235 | 2.194 | 2.276 | 2.309 | 2.427 | 2.395 | 2.456 | 2.445 | 2.344 | 2.250 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 744   | 861   | 896   | 896   | 876   | 861   | 879   | 851   | 840   | 784   | 727   | 695   | (42) |      |
| (43) |                                                                   | Edh at Noon | 68    | 86    | 110   | 130   | 126   | 123   | 108   | 106   | 90    | 75    | 62    | 58    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 0.80  | 1.74  | 2.79  | 4.32  | 5.20  | 5.31  | 6.00  | 5.18  | 3.49  | 1.92  | 0.85  | 0.63  | (44) |      |
| (45) |                                                                   | RadStd      | 0.10  | 0.20  | 0.34  | 0.38  | 0.48  | 0.52  | 0.53  | 0.35  | 0.35  | 0.26  | 0.12  | 0.09  | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (11 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page